Element B

Creating a functional and plausible heavy animal lift for the Nashville Zoo Veterinary Hospital has been a goal of Brentwood High for two years and counting. Our goal for this project was essentially to redesign some of last year's flaws in the designs while re-incorporating their successes. What this means, however, is that we spent a lengthy amount of time researching past attempts, including last year's Brentwood High engineers, and documenting ideas that we liked and believed could be utilized in our design.

| General Description | Pros | Cons | Citation | Image |
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| The UC Davis Large Animal Lift is a lightweight, simple to use, affordable piece of equipment that upholds the large horses recovering from anesthesia. Uses a balancing beam at the top of the framework | Pulls recumbent horses from stalls or trailers lifting or pulling horses stuck in mud, ravines etc. Lifting weak older horse that get down and can't rise preventing horses with pelvic injuries or other orthopedic injuries from laying down. | It utilizes straps that are designed to fit a specific species of animal (horse), rather than being useful or adjustable for a variety of species of various sizes and weights. | UC Davis Large Animal Lift – Richard Beal (richardbealblog.com) |  |
| Simplified yet heavy; used on a dairy farm to hold large cows up for varied amounts of time | Metal parts are plated while webbing and thread are made of durable polyester. Daisy Lifter weighs only 22 lbs. yet will lift animals up to 2200 lbs. The cow is gently | Again, this lift is specifically designed and produced for a single type of animal, reducing its varietal uses. | Daisy Heavy Duty Cow Lifter - Hamby Dairy Supply |  |

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| | rolled onto the sling, straps are adjusted and the C-hooks secured to a front end loader or bucket. | | | |
| The SCD09 animal lift is used to transport manatees at the Tampa Zoo to and from care pools and transfer trucks. Its structure is made with two metal beams held together with blue plastic, and is attached to a crane that controls mobilization. | Provides a practical daily weighing solution<br>Efficiently transports manatees from care pools to transfer trucks (vice versa)<br>Can hold up to 2,000 lbs<br>Manatees are easily accessible to zoo and veterinary staff<br>Uses adjustable straps that can change the height of the lift<br>Durable to water | Requires crane to lift structure; would not be eligible for lifting indoors if used the same way<br>Intended to lift manatees *specifically* to and from care pools and transportation trucks<br>Not useable for animals that dont reside in water | https://www.manitowoc.com/company/news/florida-manatees-gain-lift-shuttlelift-scd09-industrial-crane-zootampa |  |

The above designs were observed from certain organizations such as zoos, dairy farms, and veterinary hospitals. We noticed some important aspects of each design regarding how the animals were holstered in, the strap materials, the means of balancing the device, and many other small yet important features. These ideas were noted and a lot of them have returned in our current working design. The main problem we faced in observing these prior attempts, however, was the fact that the existing animal lifts are all specifically designed to fit the needs of a single animal or species. This created some difficulties in translating the ideas from old designs to our new design for the Nashville Zoo because the zoo needs a lift than is universal in nature and can be used for any type of animal that they may need to operate on. However, we also found some ideas that we really liked, such as a beam at the top of the frame to help balance the lift, which we have decided to incorporate into our design as well.

In meeting with the stakeholders, we were able to see the current lift system they have--courtesy of last year's engineering class--and we got to ask some questions about what they like or don't like about the design.

We quickly learned that they had found a couple detrimental flaws to the design, and we made sure to note them and make them a top priority in our new design. As it is shown in the top left photo, the base of the lift was connected to the lift hook by a metal wire clamp that clamps the metal wires together in a relatively central position. This in itself caused a major problem in the design because the clamp would shift to one side of the hook, causing the entire lift to lean. Additionally, as shown in the large photo on the right, the straps all connected to the tarp in an "X" shape, creating a focus point in the middle. What this means in context of the lift is that all of the animal's weight is centered at the middle of their body, causing an extremely unpleasant and hazardous--and possibly lethal--experience for the animal. We noted these as the two biggest flaws in the old design, and we made them our top priorities to improve for our redesign.

On the other hand, the old design did have successes as well; the overall design of the top frame and the utilization of straps and ratchets for the connectors were effective, and we plan on using a similar approach. They successfully allow the user to change the length of the ropes and adjust the connection points of the straps as well. Unfortunately, the top frame tended to bend and deform under heavy weight, as tested and recorded by the group last year, so we knew we would need to redesign the material and shape accordingly.

Works Cited


Beal, Richard. “UC Davis Large Animal Lift.” *richardbealblog.com*, 2 Aug. 2012, www.richardbealblog.com/uc-davis-large-animal-lift/.

Crouse, Amy. “Florida Manatees Gain ‘lift’ from Shuttlelift Scd09 Industrial Crane at Zootampa.” *Manitowoc*, 18 Aug. 2022, www.manitowoc.com/company/news/florida-manatees-gain-lift-shuttlelift-scd09-industrial-crane-zootampa.

“Daisy Heavy Duty Cow Lifter.” *Hamby Dairy Supply*, hambydairysupply.com/daisy-heavy-duty-cow-lifter/. Accessed 13 May 2024.